causing toxic chemicals. This is a commercially available product that costs less than ₹20 per filter (according to their website) and last for an average of five cigarettes – and Log 9 holds the patent to the idea (https://dgit.in/Log9Patent).

India's innovations are not limited to technologies alone. Indian startups are in the forefront when it comes to fighting superbugs. India, as a country, suffers from a massive superbug problem. According a report (https://dgit.in/SbugStat), superbugs are killing more than 7,00,000 people a year around the world. India is one of the worst affected countries (https://dgit.in/ResMap). Factors contributing to the problem include indescriminate use, abuse of antibiotics as fattening agents in the poultry industry, and a terrible record of public hygiene. India is one of the most brutal battlegrounds for the fight against superbugs. Some Indian startups are diving headfirst into that fight.

Bugworks Research India Pvt. Ltd. was founded by Anand Anandkumar after his father succumbed to an infection. The startup, based out of Bengaluru and USA, focuses on fighting Gram-Negative bacterial infections. They've developed an eflux avoidance platform that attacks bacteria in two ways at once making it harder for the bug to develop resistance. According to Anandkumar, for animals, their solution has worked against lung, blood and urinary tract infections. Human trials could follow in two years.

Post Purification Filter, or PPuF, aims to cut down the harmful effects of smoking using Graphene

bugworks

Bugworks is trying to fight against superbugs where the battlefield is the most challenging

Bugworks is not alone in this effort. A number of biotech startups in the country are trying to solve a problem that the whole world will benefit from, and which can perhaps have no better place to be solved than India.

STARTUP INDIA

Startup India is an effort by the Ministry of Commerce and Industry to cultivate and encourage the enterpreneurial ecosystem in India. Under the initiative, registered startups can enjoy an accelerated patent application process, avail tax benefits for a period of three years, apply for funds, or wind up operations in under 90 days if things do not work out. The initiative has a corpus of Rs 10,000 crore towards funding startups, but provides information on alternative funding options as well. So far, 129 startups have been funded directy.

The Startup India Hub allows for entrepreneurs to get a leg up in their ventures. Startups can look up promising incubators, find mentors, or gain access to angel investors and venture funds. The portal hosts a learning program, that is available in Hindi and English. The program consists of modules from the founders of successful Indian startups, such as Phanindra Sama of Redbus, Bhavish Aggarwal of Olacabs, and Deep Kalra of MakeMyTrip.

GOING FORWARD

A growing trend in the startup ecosystem, particularly on the investments front, is the rise of specialist venture capitalists (VCs) and funds. These are funds that are looking to help startups which possess a deep domain product, focussed on innovation and technology, and are usually backed by a VC(s) who comes with both domain expertise and entrepreneurial experience. Funds such as Endiya Partners and pi Ventures are supporting a wide range of startups like Sigtuple Labs, a startup which uses AI to analyze visual medical data. In this case, they did more than just provide the funds. While pi Ventures helped Sigtuple design the device and streamline the manufacturing process, Endiya Partners, on the other hand, aided the formation of data and strategic partnerships in the industry. The government is helping too, apart from the Startup India initiative.

SIG{(·)}TUPLE

Sigtuple Labs is a startup which uses AI to analyze visual medical data

The government has also made it easier for startups to list themselves to the public. The Bombay Stock Exchange had announced a platform for startups earlier in June. Although the platform wasn't launched on its expected launch date of 9th July, when it does launch it promises to facilitate the listing of companies in sectors like information technology, ITeS, bio-technology and life sciences, 3D printing, space technology and e-commerce. The National Stock Exchange also has a similar platform named Emerge ITP. This provides startups a definite source of investments, provided they follow the rules mentioned by the exchanges.

Indian startups are moving beyond adapting western technologies. Now, it is time for India's startups to create products and services which are answers to problems that the rest of the world has been trying to solve.